Plinthic Ferralsols

Plinthic Ferralsols are deep, strongly weathered tropical soils characterized by the presence of *plinthite* a red, iron-rich, humus-poor mixture of clay and quartz that can irreversibly harden into *ironstone* (laterite) upon prolonged exposure to drying and heating.

The term **"Plinthic"** refers to this distinctive plinthite layer, typically occurring within the upper 50–150 cm of the profile. These soils form under alternating wet and dry tropical conditions where fluctuating water tables promote iron segregation and accumulation.

They are among the most weathered soils globally, developed on old land surfaces from iron-rich parent materials such as basalt, schist, or sandstone. Despite their good drainage and structure, their fertility is low due to advanced leaching and chemical depletion.

Drainage:

- **Wet season:** Moderately to well-drained in the upper horizons but may experience temporary saturation above the plinthic horizon where perched water tables develop.
- **Dry season:** The plinthite dries and can harden irreversibly into ferricrete (ironstone), restricting root growth and water movement.
- **Landscape:** Found on gently undulating to level old plateaus, upland plains, and terraces where erosion has exposed stable weathering surfaces.

Depth:

- Generally deep (>150 cm), but **effective rooting depth** can be restricted by the plinthic or hardened ironstone layer occurring at 50–150 cm.
- Once hardened, this layer becomes impenetrable to roots and tillage equipment.

Workability:

- **When wet:** Surface horizons are friable and workable; plinthite is soft and can be shaped but should **never** be tilled when moist to avoid accelerating hardening.
- **When dry:** Ironstone layers become extremely hard; tillage is impossible where ferricrete has formed.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (5–20 cmol(+)/kg).
- **Base saturation:** Low (<35%), reflecting high leaching and dominance of sesquioxides (Fe, Al oxides).
- **Nutrients:** Nitrogen, phosphorus, potassium, and calcium are all generally low. Phosphorus fixation is very strong due to iron and aluminum oxides.
- **Organic matter:** Typically low (<2%) except where vegetation cover is dense.
- **Micronutrients:** Iron and manganese may be abundant but often unavailable due to oxidation; zinc and boron may be deficient.

pH:
Strongly acidic to moderately acidic (4.5–5.5).

Management:

- Maintain **continuous vegetative cover** (grass, mulch, agroforestry species) to prevent erosion and reduce surface heating that accelerates plinthite hardening.
- Incorporate **organic matter** regularly to improve structure, moisture retention, and nutrient availability.
- Apply **lime** to correct acidity and enhance nutrient uptake efficiency.
- Use **phosphorus fertilizers** (e.g., rock phosphate or triple superphosphate) in small, frequent doses due to high fixation potential.
- Avoid deep tillage, especially during wet periods, to prevent exposure and hardening of the plinthite.
- Prefer **shallow-rooted or perennial crops** that tolerate moderate acidity and restricted rooting depth.
- Suitable crops include cassava, oil palm, cashew, pineapple, and some pastures; where managed, maize and legumes may perform moderately well with inputs.
- Long-term land use is best suited for **agroforestry, perennial crops, or controlled grazing systems** rather than intensive annual cropping.